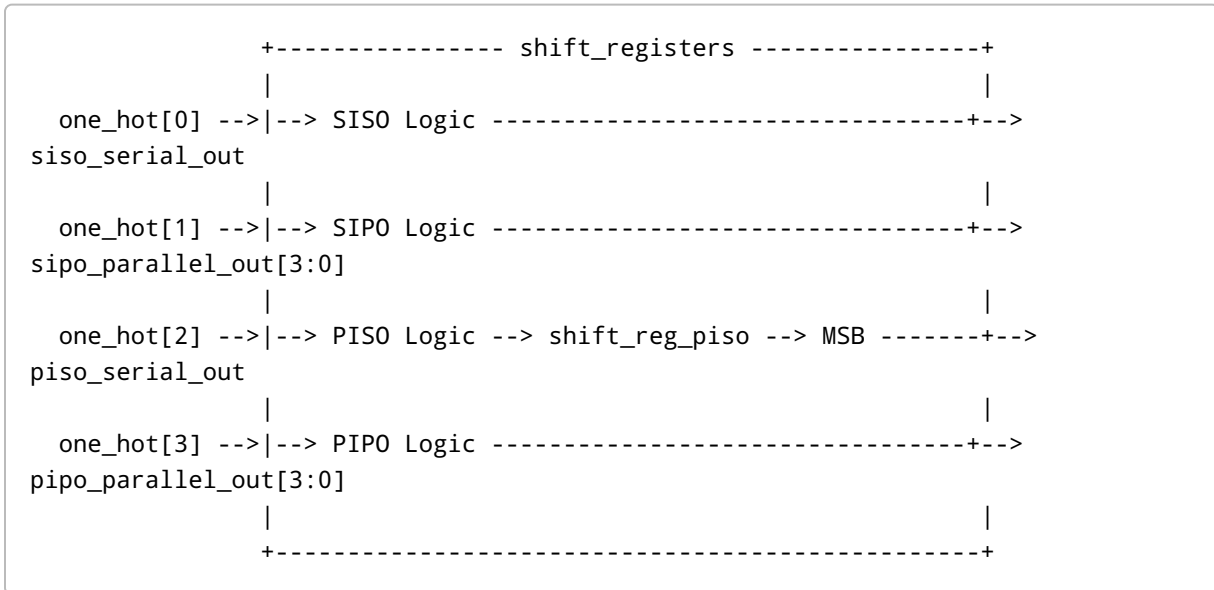


Micro-Architecture Document for `shift_registers` Module

1. Module Overview

Block Diagram (High-Level Architecture)



The `shift_registers` module implements four different types of shift register operations within a single hardware block, selected through a 4-bit one-hot encoded control signal. It supports: - **SISO** – Serial-In Serial-Out - **SIPO** – Serial-In Parallel-Out - **PISO** – Parallel-In Serial-Out - **PIPO** – Parallel-In Parallel-Out

A single shared clock and reset are used for all modes.

2. Interface Description

2.1 Common Signals

Signal	Dir	Width	Description
<code>clk</code>	Input	1	Common clock for all shift-register types
<code>rst_n</code>	Input	1	Active-low synchronous reset
<code>one_hot</code>	Input	4	Mode select: one-hot encoded

Mode Encoding: - $\text{one_hot}[0] = 1 \rightarrow \text{SISO}$ - $\text{one_hot}[1] = 1 \rightarrow \text{SIPO}$ - $\text{one_hot}[2] = 1 \rightarrow \text{PISO}$ - $\text{one_hot}[3] = 1 \rightarrow \text{PIPO}$

2.2 SISO Interface

Signal	Dir	Width	Description
<code>siso_serial_in</code>	Input	1	Serial input bit
<code>siso_serial_out</code>	Output	1	Serial output bit

2.3 SIPO Interface

Signal	Dir	Width	Description
<code>sipo_serial_in</code>	Input	1	Serial input bit
<code>sipo_parallel_out</code>	Output	4	4-bit parallel output

2.4 PISO Interface

Signal	Dir	Width	Description
<code>piso_load</code>	Input	1	Load enable: <code>1</code> = load parallel, <code>0</code> = shift mode
<code>piso_parallel_in</code>	Input	4	Parallel data to load into register
<code>piso_serial_out</code>	Output	1	MSB of shift register

2.5 PIPO Interface

Signal	Dir	Width	Description
<code>pipo_parallel_in</code>	Input	4	Parallel input
<code>pipo_parallel_out</code>	Output	4	Parallel output

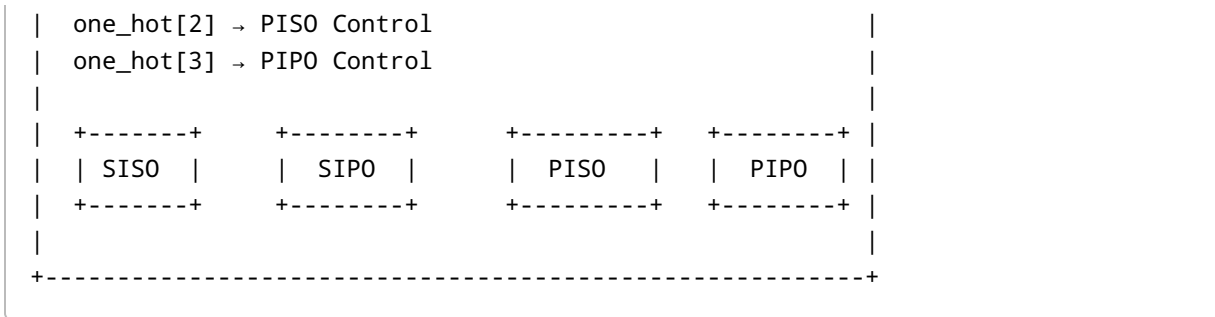
3. Internal Architecture

3.1 Block Diagram

```

+-----+
|               shift_registers Module               |
+-----+
|  one_hot[0] → SISO Control                          |
|  one_hot[1] → SIPO Control                          |
+-----+

```



Each block operates only when its corresponding one-hot bit is high. Otherwise, output registers default to 0.

4. Functional Micro-Architecture

4.1 SISO (Serial-In Serial-Out)

- On reset → output = 0
- On each clock:
- If `one_hot[0] == 1`, transfer `siso_serial_in` to `siso_serial_out`
- Else, output becomes 0

Data Flow:

```
serial_in → [FF] → serial_out
```

4.2 SIPO (Serial-In Parallel-Out)

- On reset → output = 0000
- On each clock:
- If `one_hot[1] == 1`:
 - Right-shift register
 - Insert new serial bit at LSB

```
Previous: [b3 b2 b1 b0]
New:      [b2 b1 b0 serial_in]
```

- Else → output = 0000

4.3 PISO (Parallel-In Serial-Out)

- Uses an internal 4-bit shift register `shift_reg_piso`
- On reset → `shift_reg_piso = 0`

If `one_hot[2] == 1` :- If `piso_load == 1` : load parallel data

```
shift_reg_piso = piso_parallel_in
```

- Else shift left

```
shift_reg_piso = shift_reg_piso << 1
```

Serial Output:

```
piso_serial_out = shift_reg_piso[3] // MSB
```

If mode is not active → register resets to `0000`.

4.4 PIPO (Parallel-In Parallel-Out)

- On reset → output = `0000`
 - On each clock:
 - If `one_hot[3] == 1` : transfer input directly to output
 - Else → output = `0000`
-

5. Control Logic

- All four modes are mutually exclusive via one-hot encoding.
 - No mode arbitration required.
 - Only the active mode updates its internal/output registers.
 - Inactive modes output zero (as per implementation).
-

6. Timing Behavior

Reset Behavior

- Asynchronous low-active reset
- All internal and output registers cleared

Clock Behavior

- All updates occur on rising edge (posedge clk)

7. Design Assumptions

1. One-hot signal is mutually exclusive — only one bit active at a time.
2. No simultaneous shifting and loading except as controlled by PISO logic.
3. Serial shifting direction:
4. SIPO → right shift
5. PISO → left shift

8. Limitations / Notes

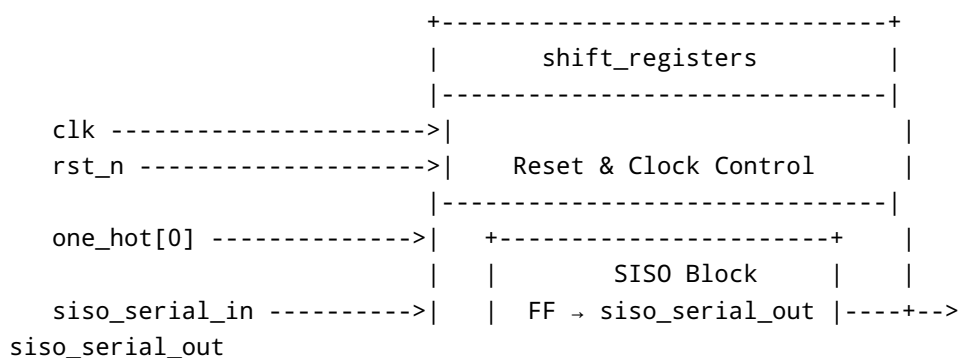
- Unselected modes output 0 (can be changed based on requirements).
- Multi-mode simultaneous operation is not supported.
- No error checking for illegal one-hot combinations.

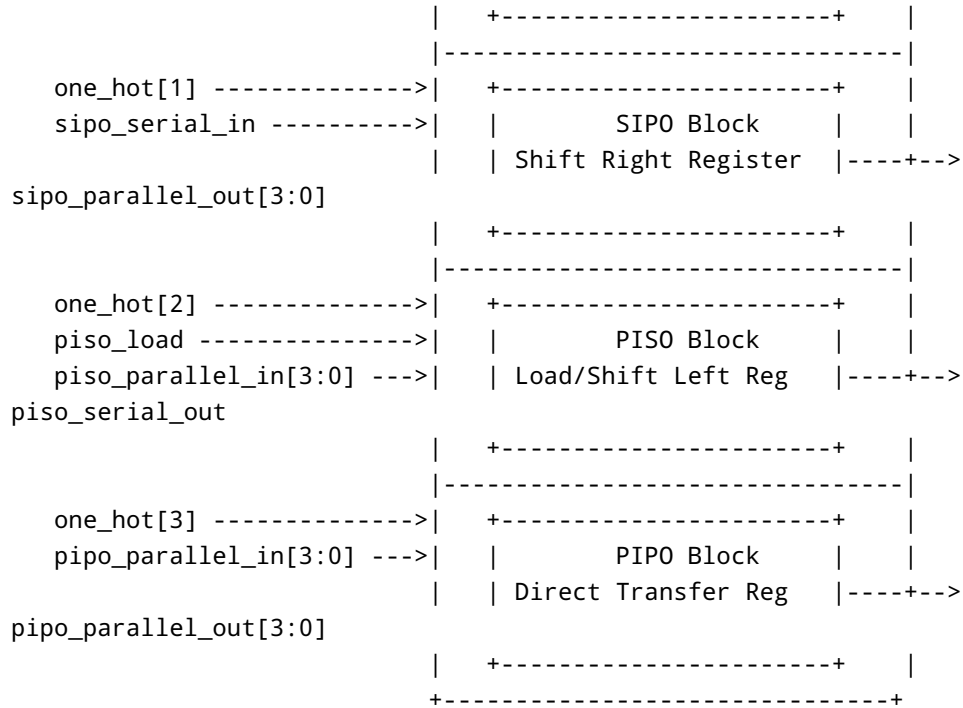
9. Future Enhancements

- Add priority logic to avoid incorrect input when multiple one-hot bits are high.
- Parameterize the width (currently fixed at 4 bits).
- Add enable signals for additional control.
- Add mode change safe-guarding to avoid glitches.

End of Micro-Architecture Document

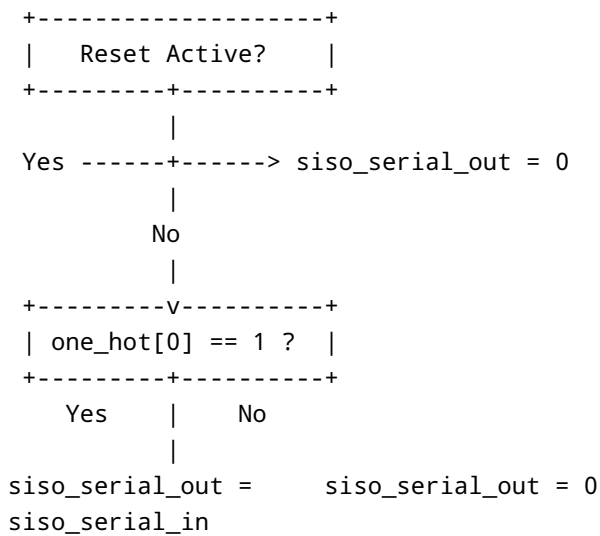
10. RTL Block Diagram (Detailed)



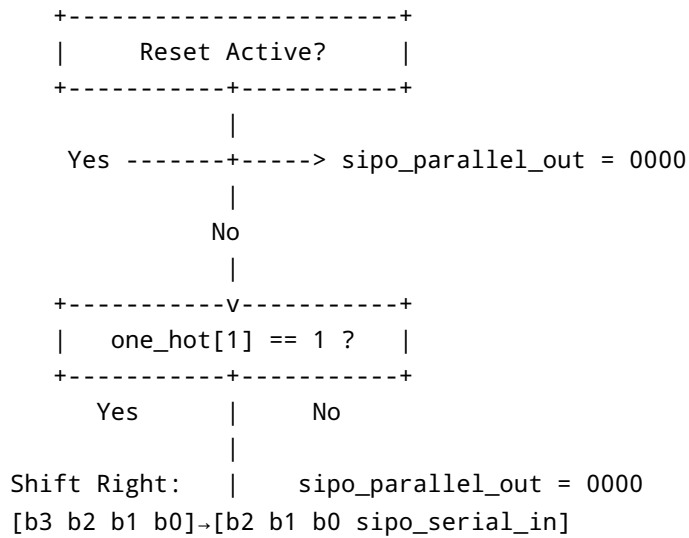


11. Flow Charts

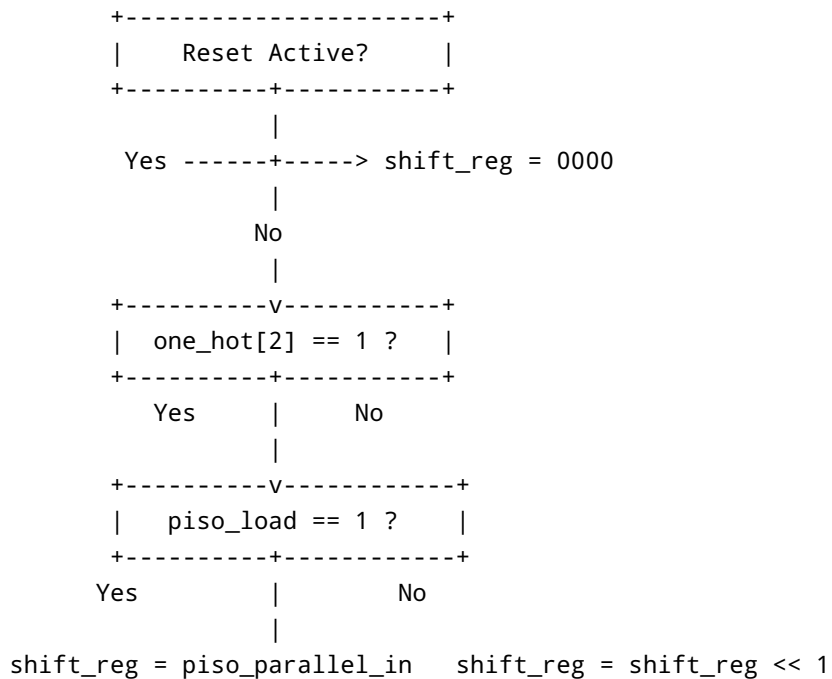
11.1 SISO Flow Chart



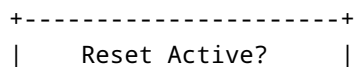
11.2 SIPO Flow Chart



11.3 PISO Flow Chart



11.4 PIPO Flow Chart



Slide 6: SIPO Operation

- Right shift register
- Collects serial bits into parallel format

Slide 7: PISO Operation

- Load parallel → shift left
- MSB becomes serial output

Slide 8: PIPO Operation

- Direct parallel transfer
- No shift operation

Slide 9: RTL Micro-Architecture

- Internal registers
- Mode decoder
- Data flow per mode

Slide 10: Flow Charts (Summary)

- SISO/SIPO/PISO/PIPO visual flow logic

Slide 11: Conclusion

- Multi-functional design
 - Supports multiple serial/parallel conversions
 - Easy integration into digital systems
-